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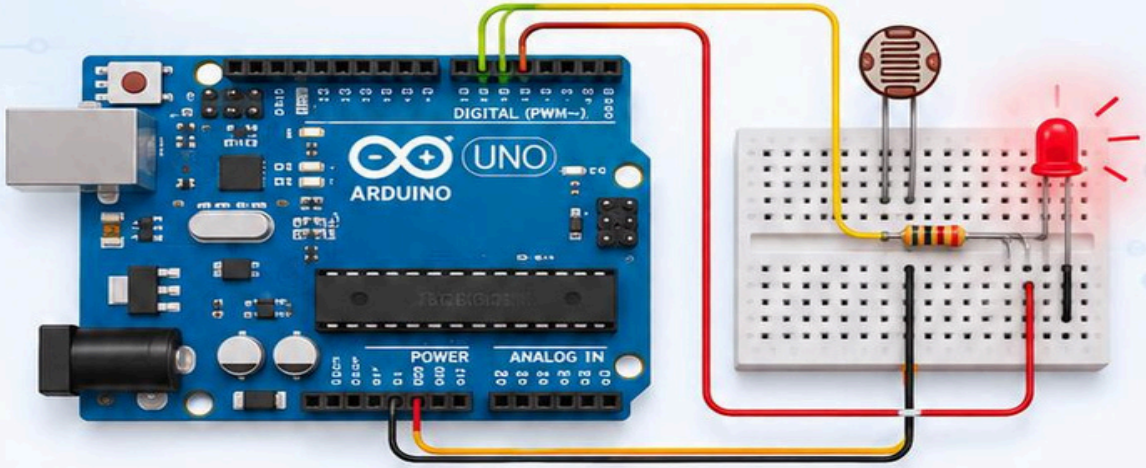
INTERNATIONAL INTERNSHIP PROGRAM

INTERNSHIP PROJECT REPORT

TASK 2 – SENSOR-BASED SIMULATION

DOMAIN: INTERNET OF THINGS (IoT)

Light Intensity-Based Automatic LED Control Using LDR Sensor



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SUBMITTED TO

CodeAlpha

Internship Program

SIMULATION PLATFORM

Tinkercad Circuits

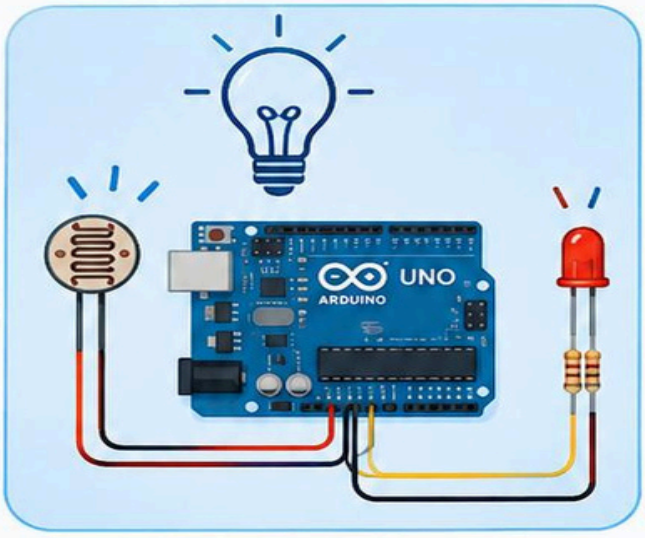
ACADEMIC YEAR: 2026 – 2027

Abstract

This project presents a simple yet effective IoT simulation that uses an LDR (Light Dependent Resistor) sensor to automatically control an LED based on ambient light intensity. The system is designed using Arduino UNO and simulated in Tinkercad Circuits. When the surrounding light is low (dark), the LED turns ON, and when the light is sufficient (bright), the LED turns OFF. This project demonstrates the basic concept of sensor-based automation in IoT, highlighting how smart devices can improve energy efficiency and enhance convenience in real-life applications.

Introduction

The Internet of Things (IoT) is a rapidly growing technology that connects physical devices to the internet, enabling them to sense, monitor and respond to real-world conditions. Sensors play a crucial role in IoT systems by detecting changes such as light, temperature, motion or distance.



An LDR (Light Dependent Resistor) is a light sensor whose resistance decreases when light intensity increases. By using an LDR with an Arduino, we can build smart systems that respond automatically to changes in the environment.

In this project, we aim to create a light-controlled LED system that turns ON in dark conditions and OFF in bright conditions. This simple project helps us understand the working of sensors, microcontrollers and their applications in smart automation.

Project Objectives

The main objectives of this project are:

- To design a simple IoT system using an **LDR** sensor and Arduino UNO.
- To automatically control an **LED** based on ambient light intensity.
- To demonstrate how sensors can be used for smart and energy-efficient solutions.
- To understand the working principle of **LDR** and Arduino in IoT applications.

Components Used

The following hardware and software components are used in this project:

Arduino UNO



Microcontroller board to process sensor data.

LDR Sensor



Detects light intensity (from dark to bright).

LED



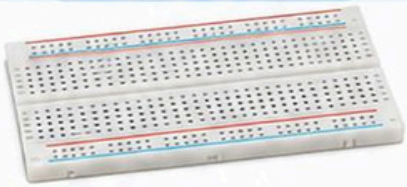
Indicates the output (ON/OFF).

Resistor



Limits current to LED (typically 220 Ω).

Breadboard



Provides a platform to make connections.

Jumper Wires



Connects all components.

Circuit Diagram and Connections

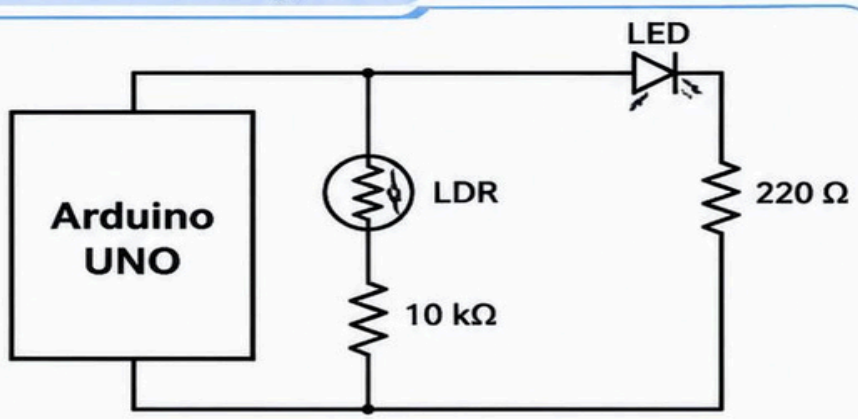
4.1 Circuit Overview

The circuit consists of an Arduino UNO, an LDR sensor, a 10 kΩ pull-down resistor, and an LED. The LDR is connected to provide analog input to the Arduino, which controls the LED based on the light intensity.

4.2 Tinkercad Circuit Screenshot



4.3 Circuit Diagram



Key Connections

- LDR connected to 5V and analog pin A0.
- 10 kΩ resistor between A0 and GND.
- LED connected to digital pin (usually D13) with a 220 Ω resistor.

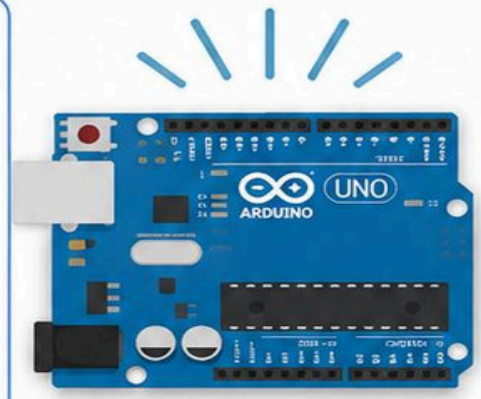
Arduino Code

```
// C++ code
//
int ldrPin = A0;
int ledPin = 8;

void setup()
{
    pinMode(ledPin, OUTPUT);
}

void loop()
{
    int lightValue = analogRead(ldrPin);

    if (lightValue < 500)
    {
        digitalWrite(ledPin, HIGH);
    }
    else
    {
        digitalWrite(ledPin, LOW);
    }
}
```





Code Explanation

1 Define Pins

ldrPin is connected to A0 (LDR sensor) and ledPin is connected to digital pin 8 (LED).

2 Setup Function

pinMode(ledPin, OUTPUT); sets pin 8 as an output pin.

3 Read Sensor Value

analogRead(ldrPin) reads the light intensity from the LDR sensor and stores it in lightValue.

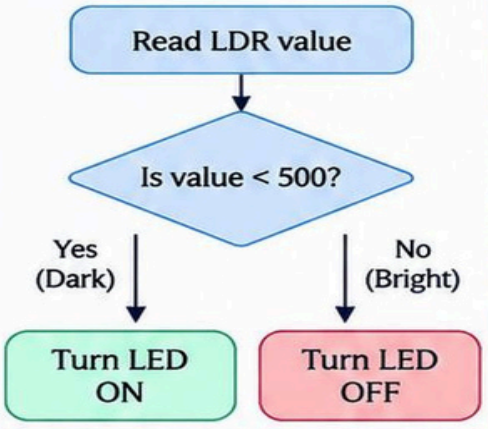
4 Check Condition

if (lightValue < 500) checks if the light is low (dark).

5 Control LED

If dark → turn LED ON (digitalWrite(ledPin, HIGH)).
Else → turn LED OFF (digitalWrite(ledPin, LOW)).

Working Logic



Simulation Results and Output

5.1 Complete Circuit in Tinkercad

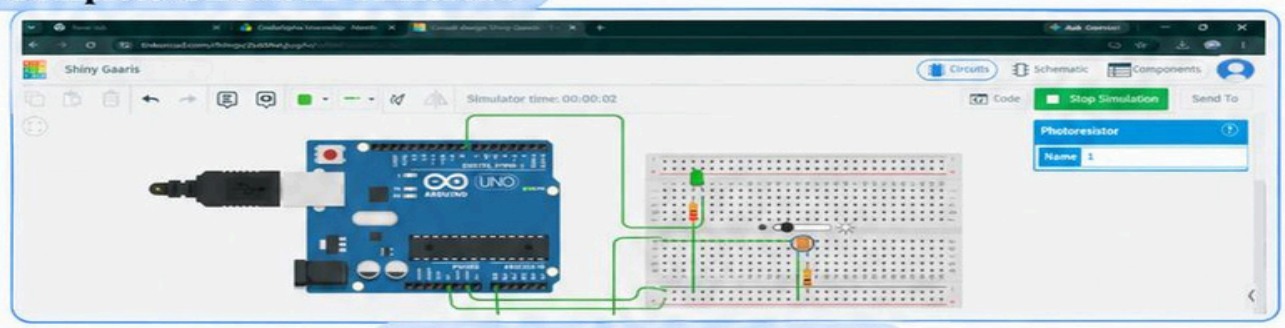


Figure 1: Complete Circuit in Tinkercad

5.2 LED Glowing in Darkness

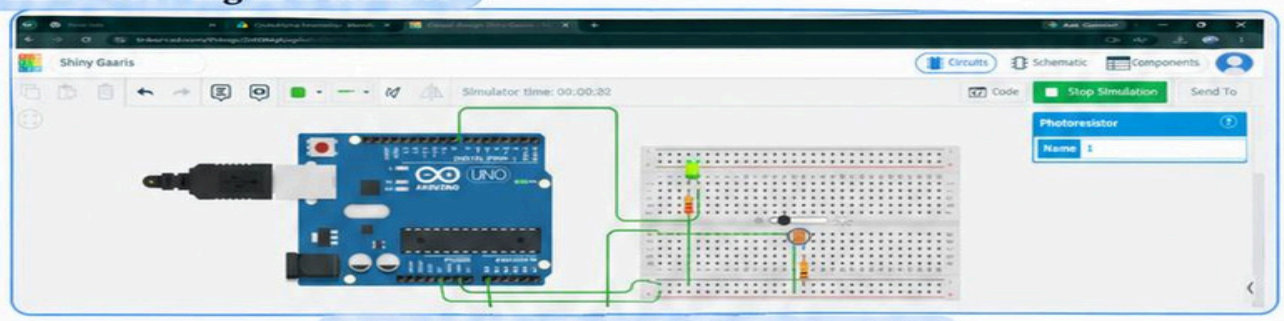


Figure 2: LED ON (Low Light – Dark Environment)

5.3 LED Off in Bright Light

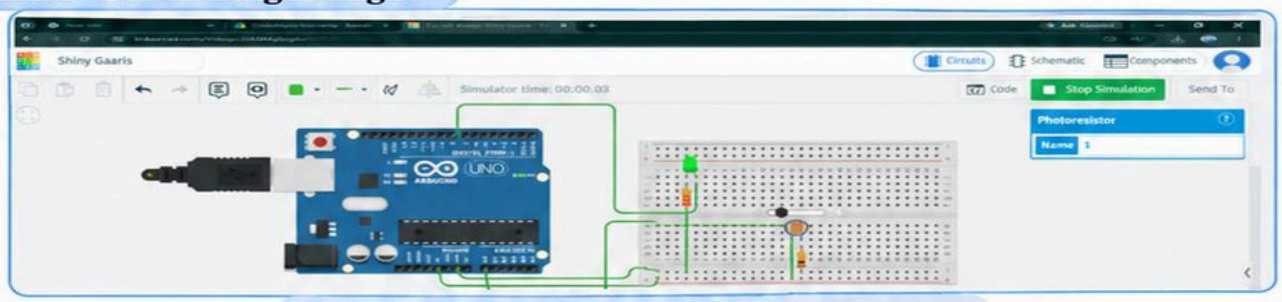


Figure 3: LED OFF (Sufficient Light – Bright Environment)

Working Demonstration

The LDR sensor detects the surrounding light. When the light is low, the LED turns ON. When light is sufficient (bright), the LED turns OFF. This demonstrates automatic light control using IoT.

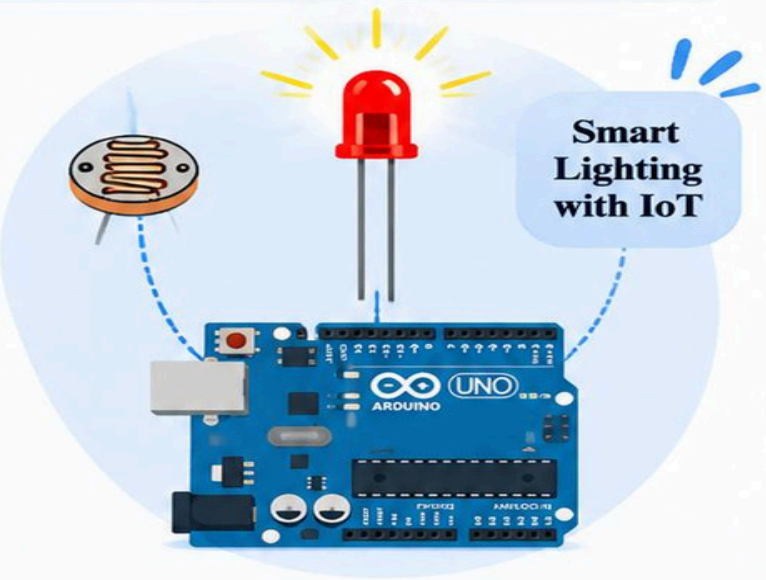
Key Observations

- LED ON in darkness
- LED OFF in bright light
- Smooth simulation in Tinkercad

6.1 Applications

This project can be used in various real-world applications, such as:

- **Automatic street lights** – turn on at night and off during daytime.
- **Smart homes & offices** – energy efficient lighting systems.
- **Security systems** – automatic lighting for safety purposes.
- **Solar-powered devices** – efficient power management.
- **IoT and embedded systems** – smart and automated solutions.



6.2 Advantages

The key advantages of this project are:

- **Energy Efficient:** Turns LED ON only when needed.
- **Low Cost:** Uses simple and inexpensive components.
- **Easy to Implement:** Built using Arduino and basic sensors.
- **Smart and Automated:** Reduces human effort.
- **Scalable:** Can be extended to larger IoT systems.

A small step towards a smarter and greener future!

7. Conclusion

This project demonstrates the effective use of LDR sensor with Arduino UNO to automatically control an LED based on ambient light intensity. By using simple components and Tinkercad simulation, we successfully designed a smart lighting system that turns ON in darkness and OFF in bright light. This project highlights the importance of IoT in creating energy-efficient, automated and intelligent solutions for everyday life. It also serves as a great foundation for learning more advanced IoT applications in the future.